

CUSTOMER CHURN PREDICTION

Simple Classification Model

Internship: CodeOrbit Tech

Role: Machine Learning Intern

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Project 2 – Classification Model

1. Project Overview

This project builds a simple classification model to predict **Customer Churn**, meaning whether a customer leaves or cancels a telecommunications service. The project follows the internship requirement to split data into training and testing sets and evaluate a classification model using accuracy, precision, and recall.

2. Objective

To train a Logistic Regression model that predicts whether a customer will stay with the service or churn, and to evaluate the model on unseen test data.

3. Dataset

The project uses the Telco Customer Churn dataset. The dataset contains 7,043 customer records and 21 original columns. The target variable is **Churn**: No indicates that the customer stayed and Yes indicates that the customer churned.

4. Data Preprocessing

The **customerID** column was removed because it is an identifier. The **TotalCharges** column was converted from text to numeric format. Eleven blank values became missing during conversion and were filled using the column median. The target was converted to numerical values (No = 0, Yes = 1). Categorical features were transformed using one-hot encoding.

5. Train-Test Split

The corrected dataset was split into 80% training data and 20% testing data using **random_state = 42**. The final workflow used 5,634 training records, 1,409 testing records, and 30 input features.

6. Classification Model

A **Logistic Regression** model was selected for this binary classification problem. The model was trained on the training set and used to predict whether customers in the test set would stay or churn.

7. Model Evaluation

The final model achieved the following results on the test dataset:

Metric	Score
Accuracy	82.11%
Precision	68.50%
Recall	60.05%

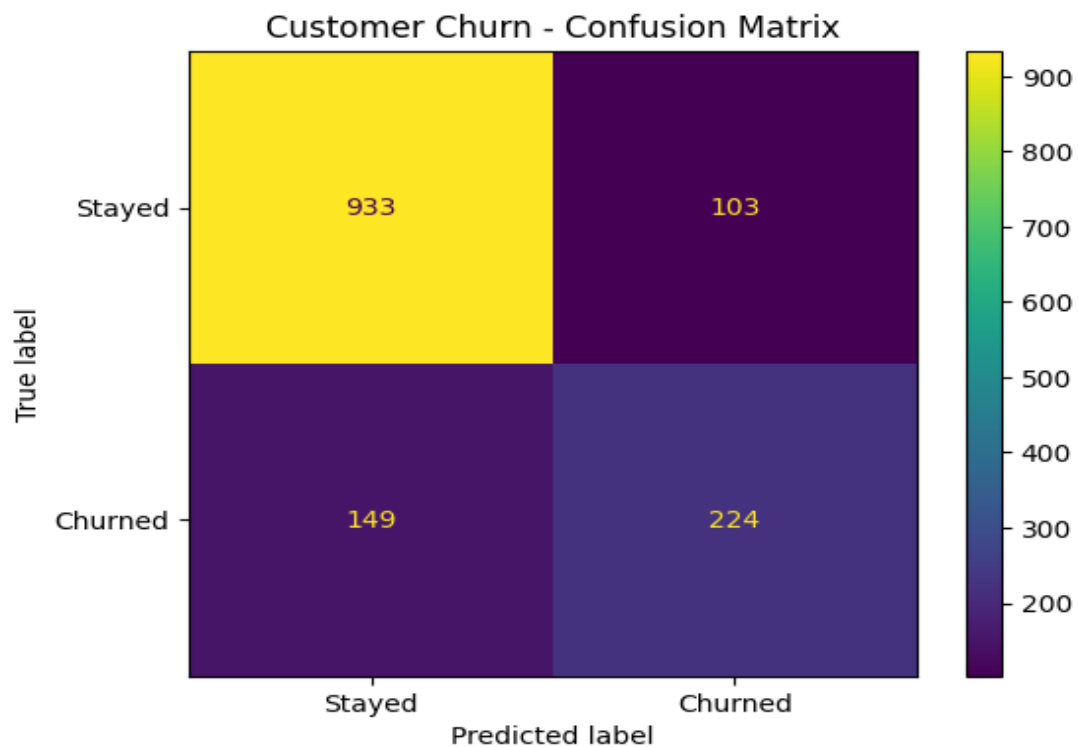
Accuracy: Measures the overall proportion of correct predictions. The model correctly classified about 82% of test cases.

Precision: For customers predicted as churned, about 69% actually churned.

Recall: The model identified about 60% of the customers who actually churned.

8. Confusion Matrix

The confusion matrix shows the model's correct and incorrect predictions for the Stayed and Churned classes.



9. Classification Report

Class	Precision	Recall	F1-score	Support
Stayed	0.86	0.90	0.88	1036
Churned	0.69	0.60	0.64	373
Overall accuracy	—	—	0.82	1409

10. Results & Interpretation

The Logistic Regression model achieved an overall accuracy of **82.11%**. It performed better at identifying customers who stayed than customers who churned. For the Churned class, the model achieved 68.50% precision and 60.05% recall, with an F1-score of 0.64. This indicates that the model can identify customer churn to a reasonable extent, while there is still scope to improve its ability to detect more customers who are likely to leave.

11. Conclusion

A Logistic Regression classification model was successfully built and evaluated for customer churn prediction. The project demonstrates the complete basic classification workflow: data preprocessing, categorical encoding, train-test splitting, model training, prediction, and evaluation using accuracy, precision, and recall.

12. Future Scope

Future improvements could include testing other classification algorithms such as Decision Tree or Random Forest, performing feature selection, tuning model parameters, and adding relevant customer-related features.

Technologies: Python, Pandas, Scikit-learn, Matplotlib, Google Colab

Final test-set performance: Accuracy 82.11% | Precision 68.50% | Recall 60.05%